

Digital Electronics

2nd Sessional Exam - Detailed Solutions

Q1. Explain in details MOSFET, RAM cell structure

Part 1: MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor)

Definition: A MOSFET is a type of field-effect transistor (FET) widely used for switching and amplifying electronic signals in electronic devices. It is the core building block of modern digital integrated circuits.

Terminals:

A MOSFET has four terminals:

1. **Source (S):** The terminal through which carriers enter the channel.
2. **Drain (D):** The terminal through which carriers leave the channel.
3. **Gate (G):** The control terminal. A voltage applied here determines the conductivity of the channel.
4. **Body/Substrate (B):** The base semiconductor material (usually internally connected to the Source).

Types of MOSFETs:

1. **Enhancement-type (E-MOSFET):** The channel does not exist naturally. A voltage must be applied to the gate to "enhance" or create the channel for current to flow. (Normally OFF).
2. **Depletion-type (D-MOSFET):** The channel exists naturally. A voltage applied to the gate depletes the channel, reducing current flow. (Normally ON).

Both types can be further divided into **N-channel** (electrons are the charge carriers) and **P-channel** (holes are the charge carriers).

Working Principle (N-channel E-MOSFET):

When a positive voltage is applied to the Gate relative to the Source ($V_{GS} > 0$), it repels holes in the p-type substrate and attracts electrons, forming an n-type conductive "channel" between the Source and Drain. When a voltage is applied between the Drain and Source (V_{DS}), current flows through this channel.

Part 2: RAM Cell Structure

A RAM (Random Access Memory) cell is the basic physical unit of memory that stores a single bit of data (0 or 1). There are two primary types of RAM cells used in digital electronics: SRAM and DRAM.

1. SRAM (Static RAM) Cell:

- **Structure:** A typical SRAM cell is made of **six transistors (6T)**. Four transistors form two cross-coupled CMOS inverters that act as a bistable latch to store the bit. The other two transistors act as access switches controlled by a Word Line (WL) to read from or write to the cell via Bit Lines (BL and BL).
- **Characteristics:** It holds data statically as long as power is supplied. It does not need to be refreshed. It is very fast but takes up more physical space on a chip. Used in CPU cache memory.

2. DRAM (Dynamic RAM) Cell:

- **Structure:** A DRAM cell is much simpler, consisting of only **one transistor and one capacitor (1T1C)**. The capacitor stores the bit of data as an electrical charge (charged = 1, discharged = 0), and the transistor acts as a switch that connects the capacitor to the bit line for reading and writing operations.
- **Characteristics:** Because capacitors leak charge over time, the data will fade unless it is periodically read and rewritten. This process is called "refreshing." DRAM is slower than SRAM but allows for much higher density (more memory in less space). Used in main system memory.

Q2. Explain Asynchronous counter with truth table

Definition:

An Asynchronous Counter (also known as a Ripple Counter) is a sequential logic circuit consisting of a series of flip-flops used to count clock pulses. The defining characteristic of an asynchronous counter is that **not all flip-flops are clocked simultaneously by the same external clock signal**.

Instead, the external clock is applied only to the first flip-flop. The output of the first flip-flop serves as the clock input for the second flip-flop, the output of the second acts as the clock for the third, and so on. The clock signal "ripples" through the circuit.

Example: 3-Bit Asynchronous Up-Counter

Let's explain a 3-bit asynchronous up-counter using JK flip-flops operating in "toggle" mode.

• Circuit Construction:

- Three JK flip-flops (FF_0, FF_1, FF_2) are used.
- The J and K inputs of all flip-flops are tied to Logic 1 (HIGH) so they toggle their output on every falling edge of their respective clock.
- The main clock is applied to the clock input of FF_0 (the Least Significant Bit, Q_0).
- The output Q_0 of FF_0 is connected to the clock input of FF_1 .
- The output Q_1 of FF_1 is connected to the clock input of FF_2 (the Most Significant Bit, Q_2).

Working:

1. Initially, all flip-flops are cleared ($Q_2 Q_1 Q_0 = 000$).
2. On the first falling edge of the main clock, FF_0 toggles, making $Q_0 = 1$. The state is 001 .
3. On the second falling edge, FF_0 toggles back to 0. The high-to-low transition of Q_0 acts as a falling edge clock for FF_1 , causing FF_1 to toggle to 1. The state is 010 .
4. This rippling effect continues, counting upwards in binary.

Truth Table (3-Bit Asynchronous Up-Counter):

Clock Pulses	Q_2 (MSB)	Q_1	Q_0 (LSB)	Decimal Value
Initial	0	0	0	0
1	0	0	1	1
2	0	1	0	2
3	0	1	1	3
4	1	0	0	4
5	1	0	1	5
6	1	1	0	6
7	1	1	1	7
8	0	0	0	0 (Recycles)

Q3. Draw & explain R S flip flop with TT

Definition:

An SR (Set-Reset) flip-flop is the most basic sequential logic circuit and acts as a 1-bit memory device. It is a bistable multivibrator, meaning it has two stable states (0 and 1). It can be constructed using cross-coupled NOR gates or NAND gates.

Below is the explanation for an Active-High SR Flip-Flop constructed using NOR gates.

Circuit Construction (Mental Draft):

The circuit consists of two NOR gates.

- The output of the first NOR gate ($\bar{Q}$) is fed back as an input to the second NOR gate.
- The output of the second NOR gate (Q) is fed back as an input to the first NOR gate.
- The other free input of the first NOR gate is designated as **R (Reset)**.
- The other free input of the second NOR gate is designated as **S (Set)**.

Inputs and Outputs:

- **S (Set):** Used to set the output Q to 1.
- **R (Reset):** Used to reset the output Q to 0.
- **Outputs:** Normal output Q and inverted output $\bar{Q}$. (By definition, $\bar{Q}$ must always be the exact opposite of Q).

Truth Table:

Here, Q_n represents the present state (before the clock/input change), and Q_{n+1} represents the next state.

S (Set)	R (Reset)	Output Q_{n+1}	State Name / Explanation
0	0	Q_n	Hold (Memory): No change. The output remains exactly what it was previously.
0	1	0	Reset: The output is forced to 0, regardless of what it was before.
1	0	1	Set: The output is forced to 1, regardless of what it was before.
1	1	Invalid/Race	Forbidden: Both outputs try to become 0 simultaneously, violating the rule that outputs must be complements.

Q4. Draw & explain briefly shift register

Definition:

A Shift Register is a type of sequential logic circuit primarily used for the storage and transfer of digital data. It is constructed by cascading a series of Flip-Flops (most commonly D-type Flip-Flops), where the output of one flip-flop is connected to the data input of the next. All flip-flops in a shift register share a common clock signal, ensuring they operate synchronously.

Working Principle:

With every active edge of the clock pulse, the data stored in the register shifts by one position (either to the left or to the right, depending on the design).

Types of Shift Registers:

Shift registers are classified based on how data is entered (Input) and how it is retrieved (Output):

1. **SISO (Serial-In, Serial-Out):** Data is entered sequentially one bit at a time, and extracted one bit at a time from the last flip-flop.
2. **SIPO (Serial-In, Parallel-Out):** Data is entered serially, but all bits can be read simultaneously from the outputs of all flip-flops.
3. **PISO (Parallel-In, Serial-Out):** All bits are loaded into the register simultaneously, but data is shifted out one bit at a time.
4. **PIPO (Parallel-In, Parallel-Out):** Data is loaded simultaneously and can be read simultaneously. It acts primarily as a temporary storage register.

Brief Explanation of a 4-Bit SISO Shift Register:

- **Circuit Construction:** Consists of four D-Flip-Flops (FF_0, FF_1, FF_2, FF_3) connected in a chain.
 - A common clock signal is given to all flip-flops.
 - The external data input (D_{in}) is applied to the D-input of FF_0 .
 - The output Q_0 of FF_0 is connected to the D-input of FF_1 . Q_1 goes to FF_2 , and Q_2 goes to FF_3 .
 - The final output (D_{out}) is taken from Q_3 .
- **Operation:** If we want to store the binary number 1011, we apply the bits sequentially to D_{in} . It will take 4 complete clock cycles for the entire 4-bit number to enter the register, and another 4 clock cycles to completely shift it out.

Applications:

- Data format conversion (Serial to Parallel communication, like in UART/USB).
- Creating time delays in digital circuits.
- Digital arithmetic operations (shifting left multiplies a binary number by 2; shifting right divides it by 2).